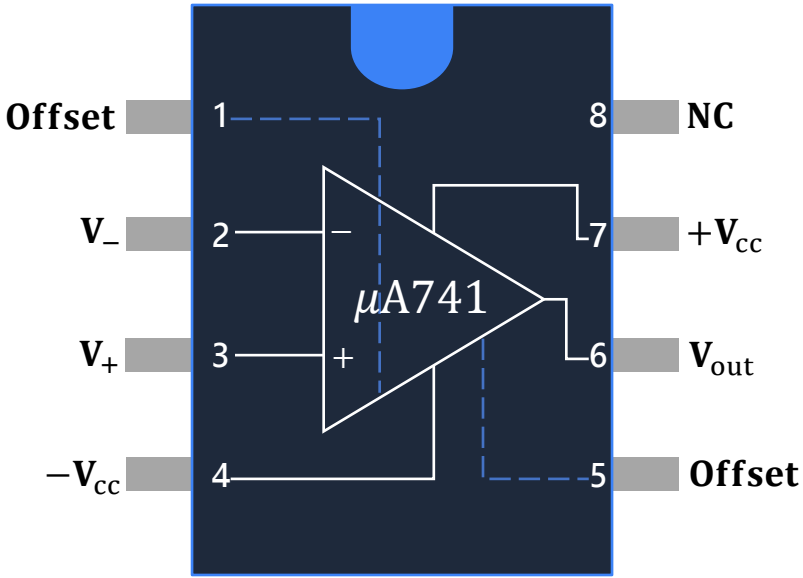
電子學實習 II：運算放大器應用電路

單元：反相放大器、非反相放大器

電子學實習授課簡報 | 成大南工 資訊科

實習教師：鄭佳榮

μA741 積體電路元件腳位定義

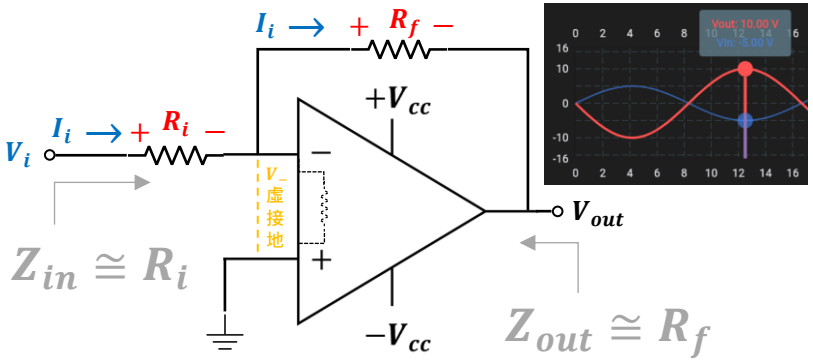


腳位	功能說明
1、5	平衡微調 (Offset Null)
2	反相輸入 (V_-)
3	非反相輸入 (V_+)
4	負電源 ($-V_{cc}$)
7	正電源 ($+V_{cc}$)
6	輸出 (V_{out})
8	空接 (NC)

※ 1 腳 與 5 腳 外接 10kΩ **可變電阻**，可微調、歸零 **輸入抵補電壓** 誤差值

放大器電路

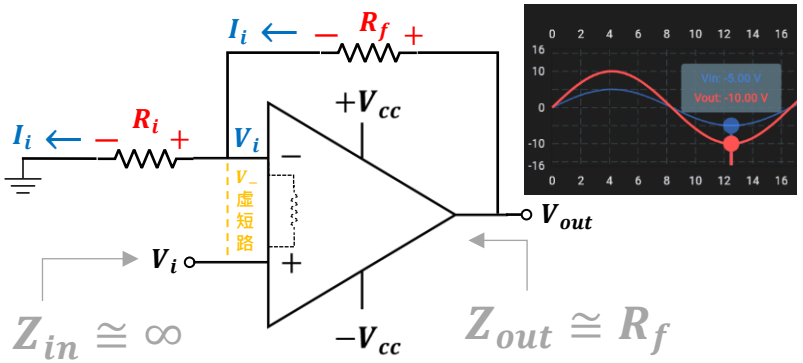
反相 放大器 (Inverting Amplifier)



$$A_{CL} = \frac{1}{\beta^-} = \frac{V_{out}}{V_i} = \frac{-I_i \times R_f}{+I_i \times R_i} = -\frac{R_f}{R_i}$$

- ✓ 輸出與輸入訊號 相位相反 (180°)
- ✓ **輸入**阻抗：受 **米勒效應** 影響 **大**，$Z_{in} \cong R_i$
- ✓ **輸出**阻抗：受 **米勒效應** 影響 **小**，$Z_{out} = R_f$

同相 放大器 (Non-Inverting Amplifier)

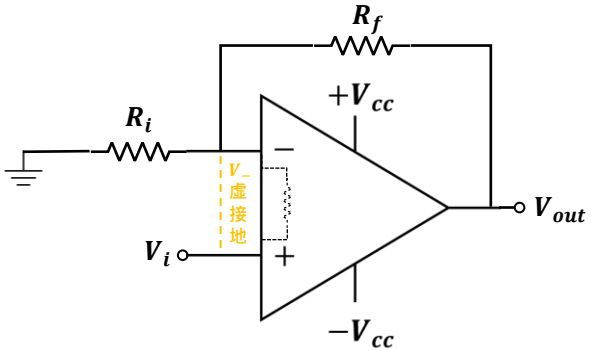


$$A_{CL} = \frac{1}{\beta^-} = \frac{V_{out}}{V_i} = \frac{V_i + (\frac{V_i}{R_i} \times R_f)}{V_i} = 1 + \frac{R_f}{R_i}$$

- ✓ 輸出與輸入訊號 同相 (0°)
- ✓ **輸入**阻抗： $Z_{in} \cong \infty$ (**極高**)
- ✓ **阻抗高入、低出** 適用 **電壓隨耦器 (緩衝器)**

放大器電路

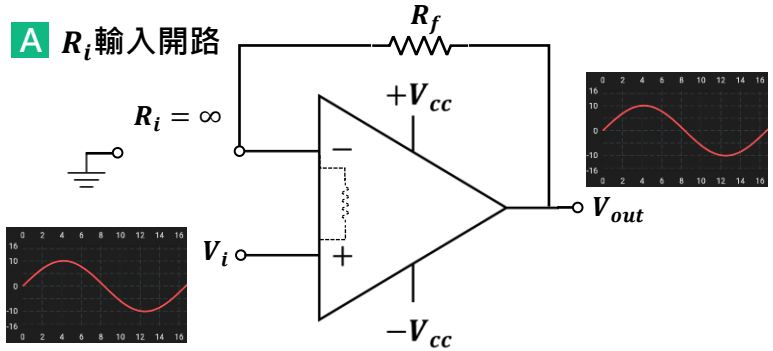
電壓隨耦器 (緩衝器)



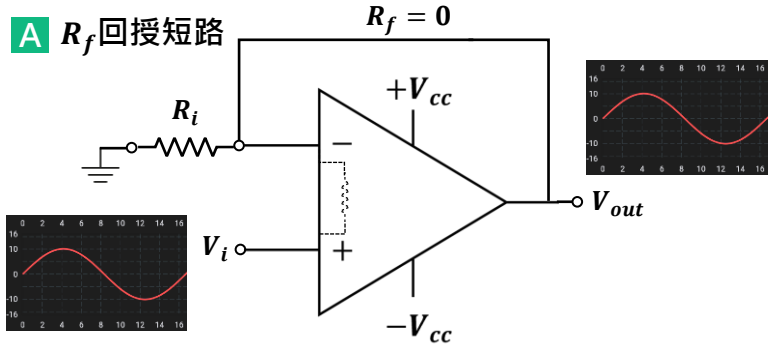
$$A_{CL} = 1 + \frac{R_f}{R_i} = 1 + \frac{R_f}{\infty\Omega\ (\text{開路})} \text{ 或 } 1 + \frac{0\,\Omega\ (\text{短路})}{R_i} \cong 1$$

- ✓ 由同相放大器組成，相位 **同相 (0°)**
- ✓ 當 $R_i = \infty\,\Omega$ 或 $R_f = 0\,\Omega$ 時，放大倍率為 1
- ✓ **高輸入**阻抗、**低輸出**阻抗，做 **緩衝器**

A R_i輸入開路

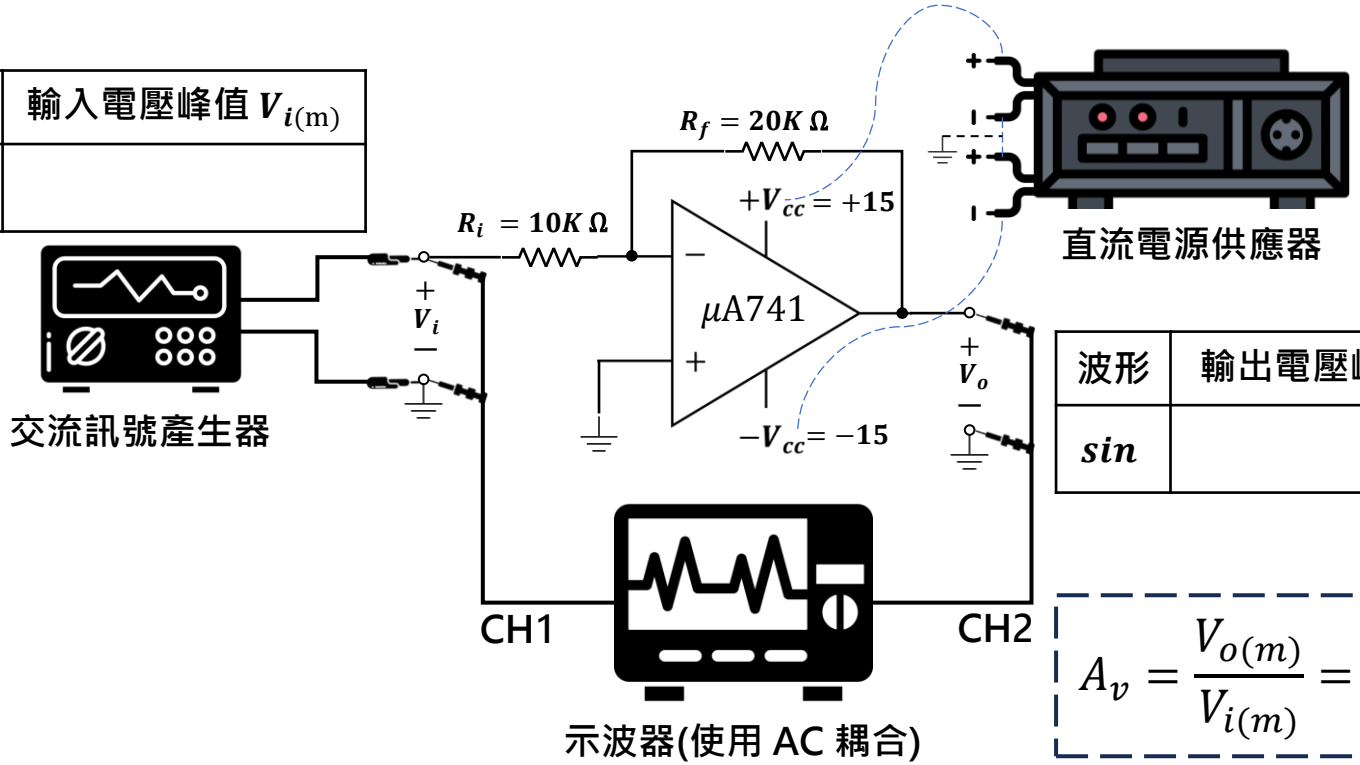


A R_f回授短路



反相放大器：電路實作 + 量測

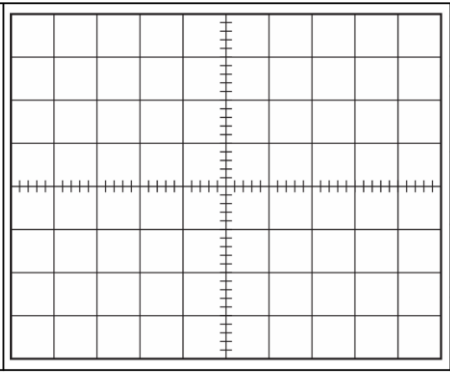
波形	輸入電壓峰值 $V_{i(\mathrm{m})}$
sin	

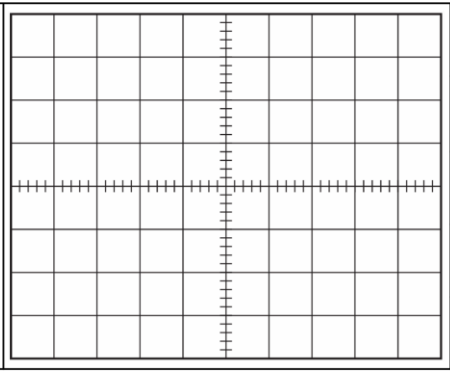


波形	輸出電壓峰值$V_{o(m)}$
sin	

$$A_v = \frac{V_{o(m)}}{V_{i(m)}} = _____$$

量測：示波器 紀錄波形 (輸入、輸出)

V_i		垂直振幅旋鈕 $VOLTS/DIV$	
		峰對峰值電壓 V_{p-p}	
		水平時間旋鈕 $TIME/DIV$	
		週期 T	
		頻率 f	

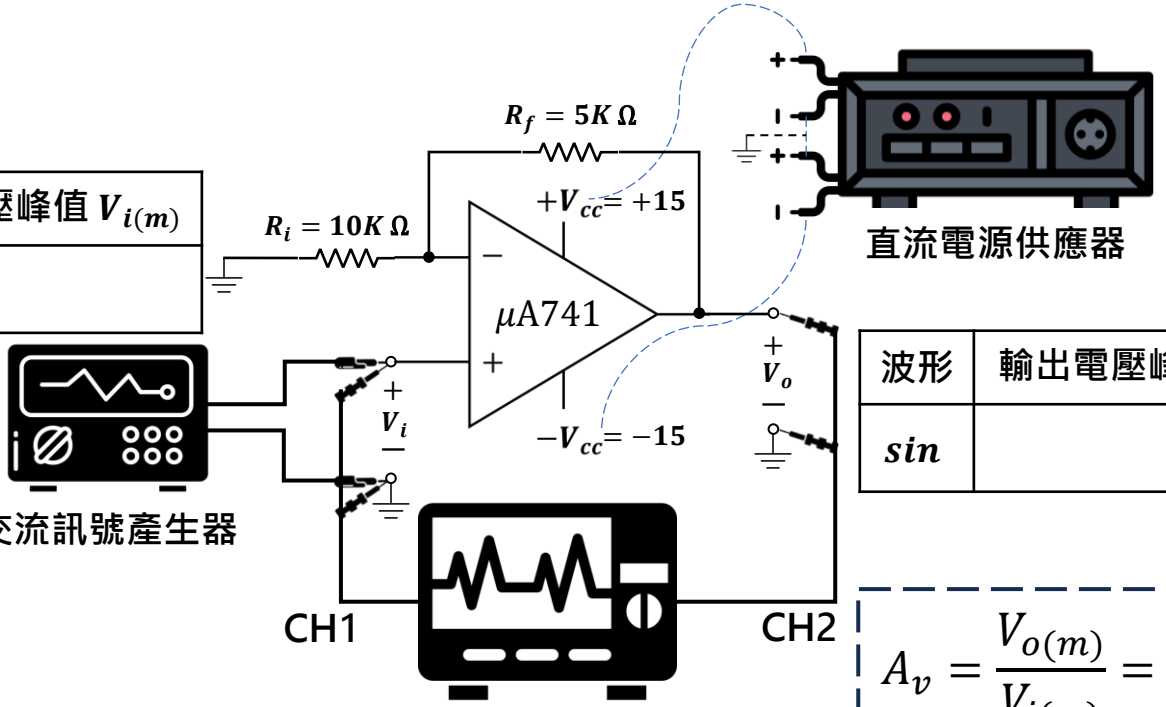
V_o		垂直振幅旋鈕 $VOLTS/DIV$	
		峰對峰值電壓 V_{p-p}	
		水平時間旋鈕 $TIME/DIV$	
		週期 T	
		頻率 f	

設計：對 R_i 與 R_f 設計數值，形成 $-\frac{1}{2}$ 倍放大器

畫出你的電路圖	

非反相放大器：電路實作 + 量測

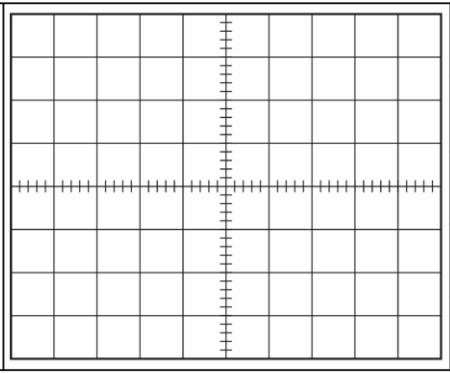
波形	輸入電壓峰值 $V_{i(m)}$
sin	

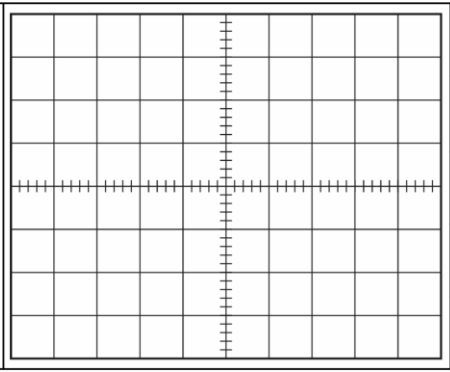


波形	輸出電壓峰值 $V_{o(m)}$
sin	

$$A_v = \frac{V_{o(m)}}{V_{i(m)}} = _____$$

量測：示波器 紀錄波形 (輸入、輸出)

V_i		垂直振幅旋鈕 $VOLTS/DIV$	
		峰對峰值電壓 V_{p-p}	
		水平時間旋鈕 $TIME/DIV$	
		週期 T	
		頻率 f	

V_o		垂直振幅旋鈕 $VOLTS/DIV$	
		峰對峰值電壓 V_{p-p}	
		水平時間旋鈕 $TIME/DIV$	
		週期 T	
		頻率 f	

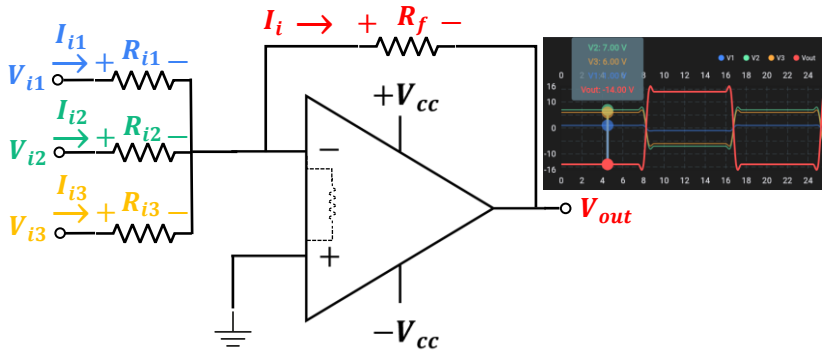
設計：對 R_i 與 R_f 設計數值，形成 1 倍放大器

畫出你的電路圖	

Thanks

加法器運算

反相 加法器 (Inverting Adder)

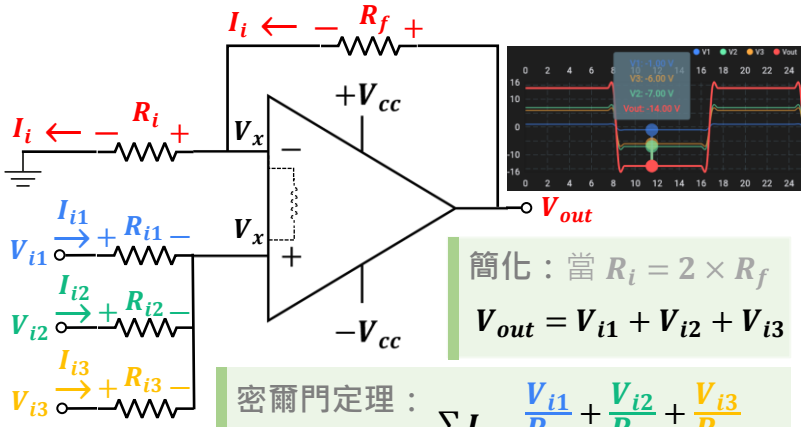


公式簡化： 當 $R_{i1} = R_{i1} = R_{i1} = \ldots = R_N = R$ 時

$$V_{out} = (V_{i1} + V_{i2} + V_{i3}) \times (-\frac{R_f}{R})$$

$$V_{out} = V_{i1} \times -\frac{R_f}{R_{i1}} + V_{i2} \times -\frac{R_f}{R_{i2}} + V_{i3} \times -\frac{R_f}{R_{i3}}$$

同相 加法器 (Non-Inverting Adder)



簡化：當 $R_i = 2 \times R_f$

$$V_{out} = V_{i1} + V_{i2} + V_{i3}$$

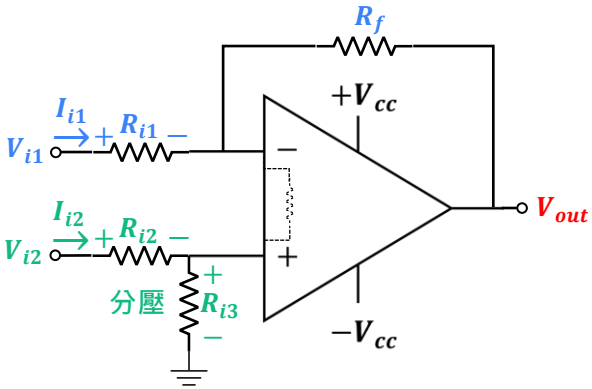
密爾門定理：

$$V_x = \frac{\sum I}{\sum G} = \frac{\frac{V_{i1}}{R_{i1}} + \frac{V_{i2}}{R_{i2}} + \frac{V_{i3}}{R_{i3}}}{\frac{1}{R_{i1}} + \frac{1}{R_{i2}} + \frac{1}{R_{i3}}}$$

$$V_{out} = V_x \times (1 + \frac{R_f}{R_i})$$

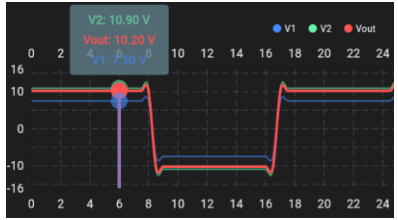
減法器運算

減法器 (Subtractor)



公式簡化 1： 當 $R_{i1} = R_f$ 且 $R_{i2} = R_{i3}$ 時

$$V_{out} = V_{i2} - V_{i1}$$



公式簡化 2： 當 $R_{i1} \times R_{i3} = R_{i2} \times R_f$ 時

$$V_{out} = (V_{i2} - V_{i1}) \times \frac{R_f}{R_{i1}}$$

重疊定理： 其中一端電壓源 短路接地

$$V_{-\to out} = V_{i1} \times -\frac{R_f}{R_{i1}}$$

$$V_{+\to out} = V_{i2} \times \frac{R_{i3}}{R_{i1} + R_{i3}} \times \left(1 + \frac{R_f}{R_{i1}}\right)$$

$$V_{out} = V_{i1} \times -\frac{R_f}{R_{i1}} + V_{i2} \times \left(\frac{R_{i3}}{R_{i2} + R_{i3}}\right) \times \left(1 + \frac{R_f}{R_{i1}}\right)$$